KNOWLEDGE

INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Department of Mechanical Engineering

TEAM NAME: **SPARKERS RACING**

GANTT CHART

TEAM MEMBERS

S no	NAME	DESIGNATION
1	Harish A	Captain
2	Karthi VS	Vice-Captain
3	Karthick SS	Driver
4	Aravindh M	Co-Driver
5	Sathivel V (HEAD)	Chasis And Fabrication
6	Madhan NG	Chasis And Fabrication
7	Karthikeyan V	Chasis And Fabrication
8	Kavinkumar M	Chasis And Fabrication
9	Prakash K	Chasis and Fabrication
10	Gokilnathan K	Chasis and Fabrication
11	Lingeshwaran KM (HEAD)	Power Train
12	Boopalan V	Power Train
13	Dharun M	Power Train
14	Gokul Ramana D	Power Train
15	Aravinth R	Power Train
16	Veeraraghavan A (HEAD)	Brake
17	Dheena S	Brake
18	Bhubesh SS	Brake
19	Harjendra DS	Brake
20	Vivin KB	Brake
21	Surendhar G (HEAD)	Steering
22	Rajendra Hari Prasth G	Steering
23	Karthikeyan V	Steering
24	Selva Ganapathy D	Steering
25	Gokuleshwaran A	Steering
26	Gowtham G	Steering
27	Guru Prasanna V (HEAD)	Design
28	Goutham R	Design
29	Gauthamganesh VS	Design
30	Deepak Sanjei SV	Design
31	Boopesh K	Design
32	Dhilipan G	Design

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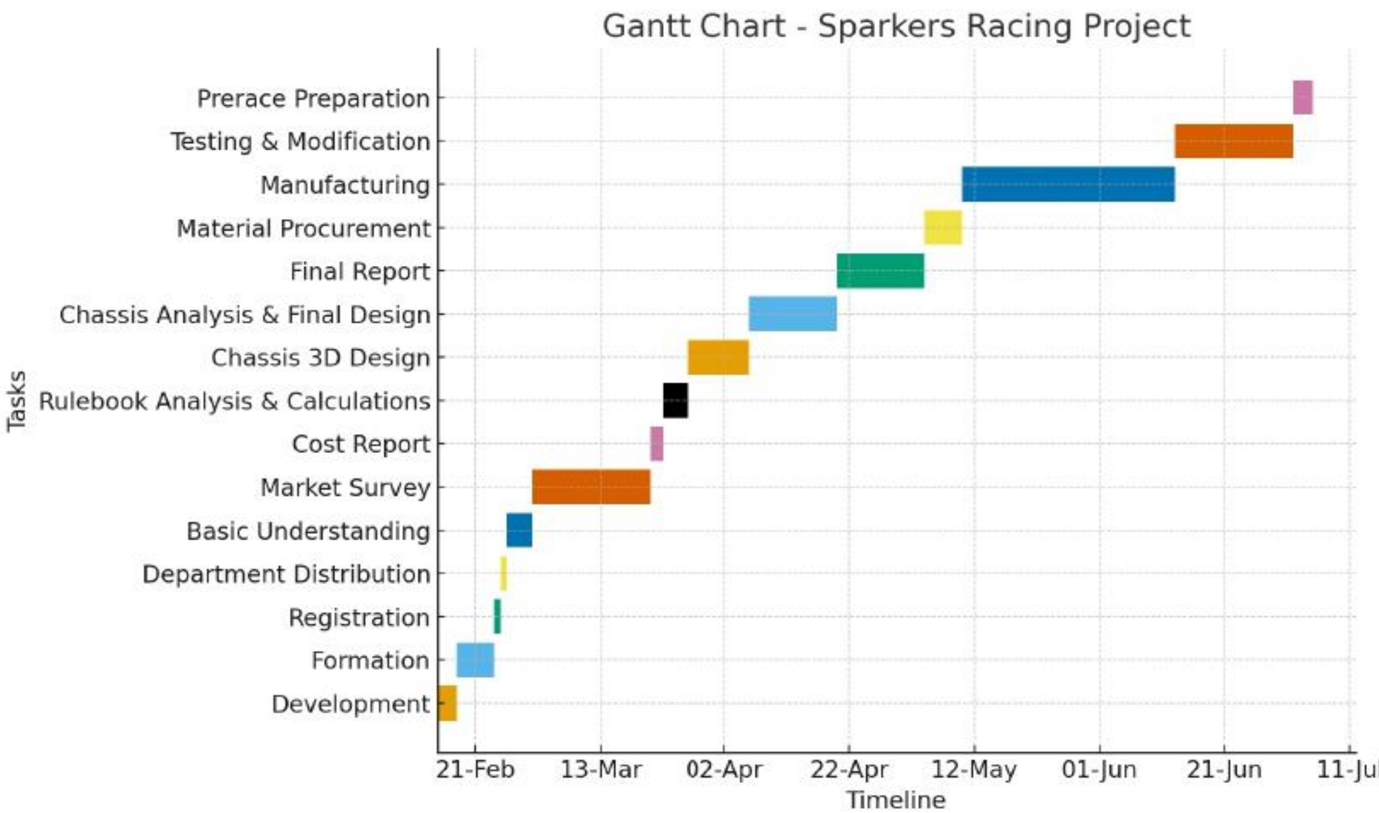
Task	Start Date	End Date
Development	15-Feb-2025	18-Feb-2025
Formation	18-Feb-2025	24-Feb-2025
Registration	24-Feb-2025	25-Feb-2025
Department Distribution	25-Feb-2025	26-Feb-2025
Basic Understanding	26-Feb-2025	02-Mar-2025
Market Survey	02-Mar-2025	21-Mar-2025
Cost Report	21-Mar-2025	23-Mar-2025
Rulebook Analysis & Calculations	23-Mar-2025	27-Mar-2025
Chassis 3D Design	27-Mar-2025	06-Apr-2025
Chassis Analysis & Final Design	06-Apr-2025	20-Apr-2025
Final Report	20-Apr-2025	04-May-2025
Material Procurement	04-May-2025	10-May-2025
Manufacturing	10-May-2025	13-Jun-2025
Testing & Modification	13-Jun-2025	02-Jul-2025
Prerace Preparation	02-Jul-2025	05-Jul-2025


Gantt Chart - Sparkers Racing Project
Prerace Preparation
Testing & Modification
Manufacturing
Material Procurement
Final Report
Chassis Analysis & Final Design
Chassis 3D Design
Rulebook Analysis & Calculations
Cost Report
Market Survey
Basic Understanding
Department Distribution
Registration
Formation
Development
Tasks
21-Feb
13-Mar
02-Apr
22-Apr
12-May
01-Jun
21-Jun
11-Jul
Timeline

1. Planning and Formation

1.1 Development (15/02/25 – 18/02/25)

In the initial stage, the main challenge was to bring together ideas and set a clear direction for the project. Under the leadership of Harish A, we worked on planning the overall framework and discussed the feasibility of completing the project within the limited time. Since it was the foundation phase, there were struggles in aligning everyone’s thoughts, but by the end of the 4 days we had a proper roadmap to proceed.

1.2 Formation (18/02/25 – 24/02/25)

Forming the team was another big step. Not everyone was confident initially about their roles, and some members were unsure about handling responsibilities. During this week, we overcame communication gaps and distributed roles among us. Once everyone was confident about their contribution, the project slowly started moving forward.

1.3 Registration (24/02/25 – 25/02/25)

The registration process was quite fast but needed accurate documentation. Even though it looked like a simple task, we had to ensure that all the details were correctly entered. After a day of effort, our registration was completed successfully, marking our official entry into the competition.

1.4 Department Distribution (25/02/25 – 26/02/25)

This was a crucial stage where the project responsibilities were formally divided. Initially, there was confusion about who should handle what, as everyone had different strengths. After some discussions and minor conflicts, the responsibilities were finalized, ensuring that each member was assigned to a suitable department.

1.5 Basic Understanding (26/02/25 – 02/03/25)

This period was all about gaining clarity. We studied previous projects, understood the scope, and clarified the rules. Though it felt overwhelming in the beginning, these five days gave us confidence to move into the next stage.

2. Research and Analysis

2.1 Market Survey (02/03/25 – 21/03/25)

The market survey was one of the longest activities. Led by Karthi VS, we struggled to collect reliable data about the availability of materials and their costs. Many vendors gave different quotations, and comparing them consumed time. But this process gave us a strong base to estimate expenses and select suitable resources.

2.2 Cost Report (21/03/25 – 23/03/25)

Based on the survey, Boopesh K prepared the cost report. The challenge here was balancing quality and budget. We debated a lot about where to cut costs and where not to compromise. Finally, within 3 days, we created a practical and efficient cost report.

2.3 Calculations (23/03/25 – 27/03/25)

This stage was handled by Guru Prasanna V. It involved complex mathematical work for performance and design. We faced difficulties in finalizing accurate values because small errors affected the overall design. After several corrections, the calculations were verified.

2.4 Rulebook Analysis (23/03/25 – 27/03/25)

Parallel to calculations, the team studied the rulebook thoroughly. At first, we felt the rules were too strict, but later we realized they guided us to avoid mistakes. Struggles came when some of our design ideas did not match the rules, and we had to modify them.

3. Design and Documentation

3.1 Chassis 3D Design (27/03/25 – 06/04/25)

This was one of the most exciting stages. Using CAD, Guru Prasanna V worked on the 3D model. The initial designs had errors in dimensions, and we had to redo them multiple times. After days of trial and error, the design was completed successfully.

3.2 Chassis Analysis & Final Design (06/04/25 – 20/04/25)

The design was tested for strength and stress. At first, our model failed in certain load conditions. We struggled to optimize it but eventually found ways to reduce weight without losing strength. This was a learning experience and gave us a finalized chassis design.

3.3 Final Report (20/04/25 – 04/05/25)

The documentation process was tiring since it required compiling every detail we had worked on. We often had to rewrite sections due to mistakes. Despite the pressure, this report became our official technical record and was completed within 15 days.

4. Material Procurement and Manufacturing

4.1 Material Procurement (04/05/25 – 10/05/25)

Sourcing materials was not easy. Some vendors delayed deliveries, and we had to search for alternatives. Karthi VS coordinated the process, and after a week of effort, all required materials were arranged.

4.2 Manufacturing (10/05/25 – 13/06/25)

This was the longest and most demanding phase. Led by Lingeshwaran KM, we worked for 40 days on fabrication, welding, and assembly. We faced several technical struggles, especially with fitting and aligning parts. At times, the work felt never-ending, but with teamwork and extra hours, the manufacturing was completed.

5. Testing and Prerace

5.1 Testing & Modification (13/06/25 – 02/07/25)

Once the vehicle was built, testing started. This was the most challenging stage because our first tests showed performance issues. Some parts had to be adjusted, and modifications were repeated multiple times. It was frustrating, but every correction improved the final output.

5.2 Prerace Preparation (From 02/07/25)

The last step was ensuring the vehicle was ready for the prerace. All systems were checked carefully, safety measures were reviewed, and final touch-ups were completed. This phase gave us pride, as all our struggles finally came together to produce a working and reliable outcome.

6. CONCLUSION

Throughout the project, we went through different stages like planning, designing, manufacturing, and testing. Each phase had its own struggles, but as a team we managed to complete the tasks within the timeline. The Gantt chart clearly showed our progress and helped us stay on track. Finally, we were able to get the vehicle ready for the prerace stage after many efforts and corrections.